

The Spectral Gravity Sub-Programme

Presentation Note 4

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Abstract

The spectral gravity sub-programme of the Cosmochrony corpus addresses a single central question: why does the effective metric $g^{\mu\nu} = 2\eta^{\mu\nu}$ satisfy the Einstein equations? Starting from the projective spectral entropy functional $\mathcal{S}_\Pi[g] = \frac{1}{2} \log \det' A_g$, the sub-programme derives the Einstein tensor as the infrared-dominant response of the renormalized metric variation, reformulates it as a spectral equilibrium condition, extends it causally to gravitational wave propagation, and provides its ultraviolet Born–Infeld completion. The logical chain is:

$$\mathcal{S}_\Pi[g] \implies \delta_g \mathcal{S}_\Pi \ni c_{\text{EH}} G_{\mu\nu} \implies G_{\mu\nu} = 8\pi G_{\text{N}} T_{\mu\nu}^{(\Pi)} \implies \text{causal completion} \implies \text{BI completion}.$$

The sub-programme is constituted by four papers: Gravity 3.0, Thermodynamics, Lorentz/CausalPropagation, and BornInfeld. The scope is the gravitational sector in the absence of matter; full coupled field equations $G_{\mu\nu} = 8\pi G_{\text{N}} T_{\mu\nu}$ with matter in Lorentzian signature remain an open deliverable.

Keywords. spectral gravity; Seeley–DeWitt expansion; Einstein tensor; spectral entropy; induced gravity; Born–Infeld completion; gravitational waves; Lorentzian completion; spectral equilibrium; non-injective projection; presentation note.

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1 Motivation and Central Question

The spectral gravity sub-programme answers the following question.

Central question. The emergent geometry sub-programme (Note 2) derives the effective Lorentzian metric $g^{\mu\nu} = 2\eta^{\mu\nu}$ as a forced consequence of the admissibility structure. It does not explain why this metric is dynamical, nor why its dynamics takes the Einstein form. *Why does the projective spectral entropy functional $\mathcal{S}_\Pi[g]$ produce the Einstein equations as its metric variation?*

More precisely: define $A_g = -\nabla_g^2$ as the minimal admissible Laplace-type operator on the effective four-dimensional manifold M , and

$$\mathcal{S}_\Pi[g] = \frac{1}{2} \log \det A_g$$

as the projective spectral entropy functional. The question is: what is the geometric content of $\delta\mathcal{S}_\Pi/\delta g^{\mu\nu}$, and in which regime does it reduce to the Einstein tensor?

The answer has four stages resolved by four papers. Gravity derives the Einstein tensor as the a_2 infrared response. Thermodynamics reformulates this as a spectral equilibrium condition equivalent to the Einstein field equations. Lorentz/CausalPropagation completes the construction causally and identifies gravitational waves as infrared probes of the pre-geometric scale. BornInfeld provides the unique ultraviolet completion of the Einstein–Hilbert sector.

Remark 1 (Scope: gravity sector only). This sub-programme closes the gravitational dynamics in the absence of matter. The coupled system $G_{\mu\nu} = 8\pi G_N T_{\mu\nu}$ with explicit matter in Lorentzian signature is an open deliverable listed in Section 8. The gauge dynamics (a_4 Yang–Mills response from the vertical variation) belongs to Note 8 (Gauge–Gravity Stratification Sub-Programme).

Remark 2 (Distinction from Note 2). Note 2 (Emergent Geometry) closes the *reconstruction* of the metric: $\Pi_q \rightarrow g^{\mu\nu} = 2\eta^{\mu\nu}$. Note 4 (Spectral Gravity) closes the *dynamics* of the metric: $\mathcal{S}_\Pi[g] \rightarrow G_{\mu\nu} = 8\pi G_N T_{\mu\nu}^{(\Pi)}$. The two notes address logically distinct questions and should not be conflated.

2 Position in the Cosmochrony Programme

The spectral gravity sub-programme takes the following inputs.

From Branch I (axiomatic primitive): the non-injective projection Π generates a spectral entropy $\mathcal{S}_\Pi > 0$ as a structural consequence of non-injectivity (ENI [5]); the Born–Infeld saturation functional and the spectral length scale ℓ_{sp} are Branch I objects.

From Note 2 (emergent geometry): the effective four-dimensional Lorentzian manifold $(M, g^{\mu\nu})$ with $g^{\mu\nu} = 2\eta^{\mu\nu}$ and the minimal admissible Laplace-type operator $A_g = -\nabla_g^2$ (Q5b–Q11).

From Note 3 (gauge structure): the admissible principal bundle $P_{G_\Pi}(M, G_\Pi)$, used in the gauge-extended functional $\mathcal{S}_\Pi[g, A]$ of Q12–Q13 (Note 8 consumer).

The outputs of the sub-programme feed directly into:

- Note 8 (Gauge–Gravity Stratification): the horizontal a_2 Einstein sector and the vertical a_4 Yang–Mills sector are derived from the same functional.
- Q13 (Einstein–Yang–Mills synthesis): the coupled $G_{\mu\nu} = 8\pi G_N T_{\mu\nu}^{\text{YM}}$ system.
- Cosmological and phenomenological applications: gravitational wave predictions, galaxy rotation, Hubble tension.

3 Logical Chain of the Sub-Programme

$$\begin{aligned}
\underbrace{\mathcal{S}_{\Pi}[g] = \frac{1}{2} \log \det' A_g}_{\text{spectral entropy}} &\implies \underbrace{\delta_g \mathcal{S}_{\Pi} \ni c_{\text{EH}} G_{\mu\nu}}_{\substack{a_2 \text{ pole} \\ \text{IR dominant}}} \implies \underbrace{G_{\mu\nu} = 8\pi G_{\text{N}} T_{\mu\nu}^{(\Pi)}}_{\text{spectral equilibrium}} \\
&\implies \underbrace{\omega^2 = c^2 k^2 - \gamma \ell_{\text{sp}}^2 k^4}_{\text{causal completion}} \implies \underbrace{\sqrt{-\det(g + \ell_{\text{sp}}^2 R)}}_{\text{BI completion}}.
\end{aligned} \tag{1}$$

The chain has four conceptually distinct stages.

1. *IR dominance of the Einstein tensor.* The renormalized metric variation $\delta \mathcal{S}_{\Pi}^{\text{ren}} / \delta g^{\mu\nu}$ decomposes into an a_2 -pole term proportional to $G_{\mu\nu}$, an a_4 -term with quadratic curvature corrections, non-local form factors, and a conformal anomaly. In the weak-curvature regime $R \ell_{\text{sp}}^2 \ll 1$, the a_2 term dominates. This is the IR hierarchy (Gravity [3]).
2. *Spectral equilibrium.* The diagonal heat kernel defines a local spectral multiplier $\beta(x)^{-1} = \beta_0^{-1} - \frac{1}{6} R(x) + O(\nabla^2 R)$. Imposing a spectral equilibrium principle (extremize $\mathcal{S}_{\Pi}$ at fixed projected matter content) yields the Einstein field equations as a variational Euler–Lagrange condition, without any assumption on causal structure (Thermodynamics [4]).
3. *Causal completion.* The Lorentzian spectral action is defined via the Schwinger–Keldysh prescription. Linearization about Minkowski spacetime gives two transverse-traceless helicity- ± 2 modes and a dispersion relation $\omega^2 = c^2 k^2 - \gamma \ell_{\text{sp}}^2 k^4$, where $\gamma = 1/(180\zeta)$ with $\zeta = O(1)$ (Lorentz/CausalPropagation [2]).
4. *UV completion.* Under conditions (C1)–(C5) and Hypothesis [H-ext], the unique admissible tensorial completion of the Einstein–Hilbert sector is the Eddington-inspired Born–Infeld action $\mathcal{S}^{\text{BI}} \propto \int [\sqrt{-\det(g + \ell_{\text{sp}}^2 R_{\mu\nu})} - \sqrt{-g}]$ (Gravity Theorem 1, conditional on [H-ext]; BornInfeld [1]).

4 Constituent Papers

4.1 Infrared Einstein response: Gravity 3.0

Gravity 3.0 [3] is the foundational paper of the sub-programme. It establishes the projective spectral entropy functional $\mathcal{S}_{\Pi}[g] = \frac{1}{2} \log \det' A_g$ for a minimal Laplace-type operator $A_g = -\nabla_g^2$ on a four-dimensional Riemannian manifold, and derives the renormalized metric variation via heat-kernel methods and zeta regularization.

Three results are established.

IR hierarchy (Theorem of §4.5; structural). The renormalized spectral stress tensor decomposes as

$$\frac{\delta \mathcal{S}_{\Pi}^{\text{ren}}}{\delta g^{\mu\nu}} = c_{\text{EH}}(\mu) G_{\mu\nu} + c_{\Lambda}(\mu) g_{\mu\nu} + \beta_{\text{loc}} B_{\mu\nu} + T_{\mu\nu}^{\text{non-loc}} + \mathcal{A}_{\mu\nu},$$

where $B_{\mu\nu}$ contains quadratic curvature terms (a_4), $T_{\mu\nu}^{\text{non-loc}}$ are non-local form factors, and $\mathcal{A}_{\mu\nu}$ is the conformal anomaly. In the regime $R \ell_{\text{sp}}^2 \ll 1$, the a_2 Einstein term is infrared-dominant.

Induced Newton constant (§5; structural). The induced coupling scales as $G_{\text{N}} \sim 16\pi^2 \ell_{\text{sp}}^2$, up to scheme-dependent factors. Newton’s constant is not a free parameter: it is set by the same spectral length ℓ_{sp} that governs the BI completion.

BI uniqueness (Theorem 1; conditional on [H-ext]). Under conditions (C1)–(C5) (spectral covariance, Einstein IR limit, algebraic BI saturation, spectral isotropy, coherence extensivity) and Hypothesis [H-ext] (admissible coherence extensivity, structural analogue of [H-color] for the

gravitational sector), the unique admissible tensorial UV completion is the Eddington-inspired BI action. Lemmas 2 and 3 (scalar BI reduction and spectral carrier identification) are proved unconditionally; only Lemma 1 (coherence extensivity) is conditional on [H-ext].

Structural necessity of non-injective projection (§7.2; structural). Any genuine emergence with $\mathcal{S}_\Pi > 0$ requires non-injective projection. The gravitational response is therefore structurally tied to the non-injectivity of Π and cannot arise in a purely injective effective description.

4.2 Spectral equilibrium and the Einstein equation: Thermodynamics

Thermodynamics [4] develops the local thermodynamic interpretation of $\mathcal{S}_\Pi[g]$ and derives the Einstein field equations as a spectral equilibrium condition.

Three results are established.

Local spectral multiplier field (§2; structural). The diagonal heat kernel $K(x, x; t)$ defines a local spectral energy density $u(x; t) = -\partial_t \log K(x, x; t)$, whose Seeley–DeWitt expansion gives the local multiplier:

$$\beta(x)^{-1} = \beta_0^{-1} - \frac{1}{6}R(x) + O(\nabla^2 R).$$

Here $\beta_0^{-1} = 2/t_*$ is a universal UV contribution; the $-\frac{1}{6}R(x)$ correction is the same a_2 coefficient that produces the Einstein tensor in Gravity.

Local spectral first law (§3; structural). The compatibility between the geometric and matter sides of the metric variation gives $\delta\mathcal{S}_\Pi = \int \beta(x)^{-1} \delta E_\Pi(x)$. This is not a postulate but a consequence of the Seeley–DeWitt hierarchy.

Einstein equation as spectral equilibrium (Theorem 4.3; structural). The spectral equilibrium principle: admissible geometries extremize $\mathcal{S}_\Pi$ at fixed projected matter content $W_\Pi[g, \psi]$. The Euler–Lagrange equation is

$$G_{\mu\nu} = 8\pi G_N T_{\mu\nu}^{(\Pi)},$$

with the factor $8\pi G_N$ fixed by the product of two universal normalizations. No horizon structure, no Rindler wedge, and no Raychaudhuri theorem are required. The derivation is Riemannian; its extension to Lorentzian signature is the content of the Lorentz/CausalPropagation paper.

The relation to Jacobson’s derivation is established in §6: the present framework is structurally parallel but replaces area entropy by $\mathcal{S}_\Pi$, Rindler temperature by $\beta(x)^{-1}$, and horizon equilibrium by variational extremum. The derivations are logically independent.

4.3 Causal completion and gravitational waves: Lorentz/CausalPropagation

Lorentz/CausalPropagation [2] constructs the Lorentzian completion of the spectral entropy framework and establishes the causal propagation of gravitational degrees of freedom.

Three results are established.

Lorentzian spectral action (§2; structural). The Lorentzian spectral action is defined via the real part of the hyperbolic determinant, equivalently through a Schwinger–Keldysh prescription. This ensures a real and retarded quadratic kernel without additional assumptions.

Graviton content (§3; structural). Linearization about Minkowski spacetime in the regime $R\ell_{\text{sp}}^2 \ll 1$ and $\omega \ll \ell_{\text{sp}}^{-1}$ propagates exactly two transverse-traceless tensor modes of helicity ± 2 . Scalar and vector sectors remain non-dynamical in the infrared. The polarization content coincides with that of general relativity in the accessible domain.

Dispersion relation and k^4 correction (§3; structural). The ratio of Seeley–DeWitt coefficients fixes the universal correction:

$$\omega^2 = c^2 k^2 - \gamma \ell_{\text{sp}}^2 k^4 + O(k^6), \quad \gamma = \frac{1}{180\zeta},$$

with $\gamma = 1/30$ in the natural spectral scheme. The additional pole at $k^2 \sim \ell_{\text{sp}}^{-2}$ lies beyond current observational reach. This structure maps onto the parametrizations used in gravitational-wave catalogues and yields a bound on the pre-geometric scale ℓ_{sp} .

4.4 Born–Infeld UV completion: BornInfeld

BornInfeld [1] establishes the Born–Infeld structure as the unique gravitational UV completion compatible with bounded-flux propagation and admissible coherence extensivity.

Its main contribution to the sub-programme is the scalar BI uniqueness result: the saturation bound $|\partial_t \chi| \leq c_{\text{BI}}$ selects a square-root action density in the one-dimensional sector. Combined with Gravity Theorem 1, the tensorial extension gives the Eddington-inspired BI form.

The parity involution $\chi \mapsto -\chi$ (Lemma 2.1 of BornInfeld) is a direct structural consequence: BI-indiscernibility forces the minimal fibre condition $\Pi^{-1}(y) \supseteq \{\chi, -\chi\}$. This result feeds directly into the O-series (O18) and the parity involution $c \leftrightarrow q - c$ of the pair observable.

Table 1: Summary of constituent papers by stage. Status: P = proved; S = structural; C = conditional on [H-ext].

Stage	Paper	Central output	Status
IR Einstein response	Gravity 3.0	$\delta_g \mathcal{S}_\Pi \ni c_{\text{EH}} G_{\mu\nu}$; IR hierarchy	S
Induced G_{N}	Gravity 3.0	$G_{\text{N}} \sim 16\pi^2 \ell_{\text{sp}}^2$	S
BI scalar uniqueness	BornInfeld	Scalar BI; parity involution	P
BI tensorial uniqueness	Gravity Theorem 1	Eddington-BI form	C ([H-ext])
Spectral multiplier field	Thermodynamics	$\beta(x)^{-1} = \beta_0^{-1} - \frac{1}{6}R$	S
Spectral first law	Thermodynamics	$\delta \mathcal{S}_\Pi = \int \beta^{-1} \delta E_\Pi$	S
Einstein as equilibrium	Thermodynamics	$G_{\mu\nu} = 8\pi G_{\text{N}} T_{\mu\nu}^{(\Pi)}$	S
Lorentzian completion	Lorentz/CausalProp	Real retarded kernel; Schwinger–Keldysh	S
Helicity- ± 2 modes	Lorentz/CausalProp	Two TT modes; no scalar in IR	S
k^4 correction	Lorentz/CausalProp	$\gamma = 1/30$; bound on ℓ_{sp}	S

5 Inputs from Upstream Results

1. **Non-injectivity as structural necessity (ENI [5]).** Any genuine emergence with $\mathcal{S}_\Pi > 0$ requires non-injective projection. The existence of the spectral entropy functional is a consequence of this structural necessity, not an independent postulate.
2. **Effective four-dimensional manifold $(M, g^{\mu\nu})$ (Note 2).** The sub-programme assumes the effective base manifold derived in Q5b–Q11 and the induced metric $g^{\mu\nu} = 2\eta^{\mu\nu}$ as the starting point for the Laplace-type operator A_g . Without Note 2, the operator A_g would lack a canonical definition.
3. **Spectral length scale ℓ_{sp} (Branch I).** The scale ℓ_{sp} is the admissibility cutoff that appears in the induced Newton constant $G_{\text{N}} \sim 16\pi^2 \ell_{\text{sp}}^2$ and in the BI completion. Its origin is in the Born–Infeld saturation constant c_{BI} of the projection dynamics.
4. **BI parity involution (BornInfeld, O18).** The minimal fibre condition $\Pi^{-1}(y) \supseteq \{\chi, -\chi\}$ established in BornInfeld [1] and O18 is used in the Gravity uniqueness argument to fix the sign of the spectral carrier.

Table 2: Principal outputs and downstream consumers.

Output	Consumer	Status
$G_{\mu\nu} = 8\pi G_N T_{\mu\nu}^{(\Pi)}$ (matter-free IR)	Q13, Q6b	S
$G_N \sim 16\pi^2 \ell_{\text{sp}}^2$	Q12, Q13	S
Eddington-BI action	Q13 (EBI joint functional)	C ([H-ext])
Helicity- ± 2 gravitons	Cosmology applications	S
$\omega^2 = c^2 k^2 - \gamma \ell_{\text{sp}}^2 k^4$ dispersion	GW phenomenology	S
Bound on ℓ_{sp} from GW data	Phenomenological notes	S
BI scalar uniqueness; parity involution	O18, O30	P
$a_2 \rightarrow$ gravity; $a_4 \rightarrow$ gauge (stratification)	Note 8, Q12, Q13	S

6 Outputs to Downstream Branches

7 Status of Results

7.1 Proved results

1. *Scalar BI uniqueness* (BornInfeld [1]): the saturation bound $|\partial_t \chi| \leq c_{\text{BI}}$ selects a unique square-root action in the one-dimensional sector.
2. *BI parity involution* (BornInfeld, Lemma 2.1): $\mathcal{S}_{\text{BI}}[-\chi] = \mathcal{S}_{\text{BI}}[\chi]$; BI-indiscernibility forces the minimal fibre $\Pi^{-1}(y) \supseteq \{\chi, -\chi\}$.
3. *Spectral carrier identification* (Gravity Lemma 3; unconditional): the unique two-derivative covariant endomorphism compatible with spectral covariance and isotropy is $X = \ell_{\text{sp}}^2 g^{-1} R$ with $R_{\mu\nu}$ the Ricci tensor ($\alpha = 0$).
4. *Scalar BI reduction* (Gravity Lemma 2; unconditional): the eigenvalue function reduces to $h(\lambda) = \frac{1}{2} \log(1 + \lambda)$, forced by the scalar BI uniqueness result of BornInfeld.

7.2 Structural results

1. *IR hierarchy and IR dominance of $G_{\mu\nu}$* (Gravity §4.5): the a_2 Einstein term dominates in the regime $R\ell_{\text{sp}}^2 \ll 1$. The result is a structural statement about the renormalized spectral stress tensor; it is not an exact derivation valid at all curvature scales.
2. *Induced Newton constant* (Gravity §5): $G_N \sim 16\pi^2 \ell_{\text{sp}}^2$, up to scheme-dependent factors.
3. *Spectral multiplier field and first law* (Thermodynamics §2–3): $\beta(x)^{-1}$ is defined from spectral data; the first law $\delta \mathcal{S}_{\Pi} = \int \beta^{-1} \delta E_{\Pi}$ is a structural consequence of the Seeley–DeWitt hierarchy.
4. *Einstein equation as spectral equilibrium* (Thermodynamics Theorem 4.3): the Euler–Lagrange condition yields $G_{\mu\nu} = 8\pi G_N T_{\mu\nu}^{(\Pi)}$ in the IR. No causal structure is assumed; the derivation is Riemannian.
5. *Lorentzian completion and Schwinger–Keldysh prescription* (Lorentz/CausalPropagation §2): the real retarded kernel is defined; the spectral action is real and causal.

6. *Two helicity- ± 2 graviton modes* (Lorentz/CausalPropagation §3): scalar and vector sectors are non-dynamical in IR; polarization content matches general relativity.
7. *k^4 dispersion correction* (Lorentz/CausalPropagation §3): $\gamma = 1/(180\zeta)$; $\gamma = 1/30$ in the natural spectral scheme.
8. *Tensorial BI uniqueness conditional on [H-ext]* (Gravity Theorem 1): the Eddington-inspired BI form is the unique admissible completion. Conditionality is sharply localised to Lemma 1 (coherence extensivity from [H-ext]); Lemmas 2 and 3 are unconditional.

7.3 Open hypotheses

1. *[H-ext] (admissible coherence extensivity)*. The derivation of [H-ext] from Axiom A3 (non-factorisability) is structurally motivated but not yet analytically closed. [H-ext] occupies an analogous status to [H-color] in the spectral admissibility sub-programme: the result it conditions (BI uniqueness) is the correct structural output; the hypothesis localises the remaining analytical gap.
2. *Full Lorentzian signature of the Einstein equilibrium condition*. Thermodynamics Theorem 4.3 is proved in Riemannian signature. The analytic continuation of $K(x, x; t)$ and of the first law to the Lorentzian setting (beyond the linearized Lorentz/CausalPropagation result) is open.
3. *Coupled field equations with matter*. The spectral stress tensor $T_{\mu\nu}^{(\Pi)}$ is defined formally from the projected matter functional $W_{\Pi}[g, \psi]$. Its explicit construction in terms of standard matter fields, and the derivation of the full coupled $G_{\mu\nu} = 8\pi G_N T_{\mu\nu}$ in Lorentzian signature, remain open.

8 Open Deliverables

Three open items define the boundary of the sub-programme.

Open deliverable 1: Analytical proof of [H-ext]. This is the sole conditionality of the BI uniqueness result. Resolving [H-ext] would promote Gravity Theorem 1 to an unconditional theorem.

Open deliverable 2: Full Lorentzian Einstein equilibrium. The spectral equilibrium derivation of $G_{\mu\nu} = 8\pi G_N T_{\mu\nu}^{(\Pi)}$ is currently Riemannian. A fully covariant Lorentzian formulation, integrating the Schwinger–Keldysh prescription of Lorentz/CausalPropagation with the spectral equilibrium principle of Thermodynamics, is an open structural programme.

Open deliverable 3: Coupled field equations with matter in Lorentzian signature. The full system $G_{\mu\nu} = 8\pi G_N T_{\mu\nu}$ with explicit matter fields and without the current-free restriction is the primary open deliverable of the gravitational sector. It requires identifying the admissible projected sector for each matter species, a task that awaits the completion of Note 6 (Fermionic Matter Sub-Programme). This deliverable separates into a closed downstream part and an open upstream part. The sourced vector sector is structurally fixed once a projected momentum density $T_{0i}^{(\Pi)}$ is supplied: the linearized Lorentzian constraint produces the gravito-magnetic component g_{0i} in the standard infrared Einstein regime, reproducing frame dragging without an additional propagating degree of freedom (Lorentz/CausalPropagation [2]). The open part is upstream, namely the derivation of $T_{0i}^{(\Pi)}$ from moving projected matter configurations, which remains to be completed in the fermionic and coupled-matter sector.

9 Conclusion

The spectral gravity sub-programme derives gravitational dynamics from the projective spectral entropy functional $\mathcal{S}_\Pi[g] = \frac{1}{2} \log \det' A_g$.

The Einstein tensor arises as the a_2 infrared-dominant response of the renormalized metric variation (Gravity 3.0), reformulated without causal assumptions as the Euler–Lagrange condition of a spectral equilibrium principle (Thermodynamics). Two transverse-traceless helicity- ± 2 graviton modes propagate causally in the Lorentzian completion, with a universal k^4 correction suppressed by ℓ_{sp}^2 (Lorentz/CausalPropagation). The unique ultraviolet completion is the Eddington-inspired Born–Infeld action, conditional on [H-ext] (Gravity Theorem 1, BornInfeld).

The sub-programme does not address Yang–Mills dynamics or the gauge–gravity coupling ratio; those belong to Note 8. The full coupled Einstein field equations with matter in Lorentzian signature remain the primary open deliverable.

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